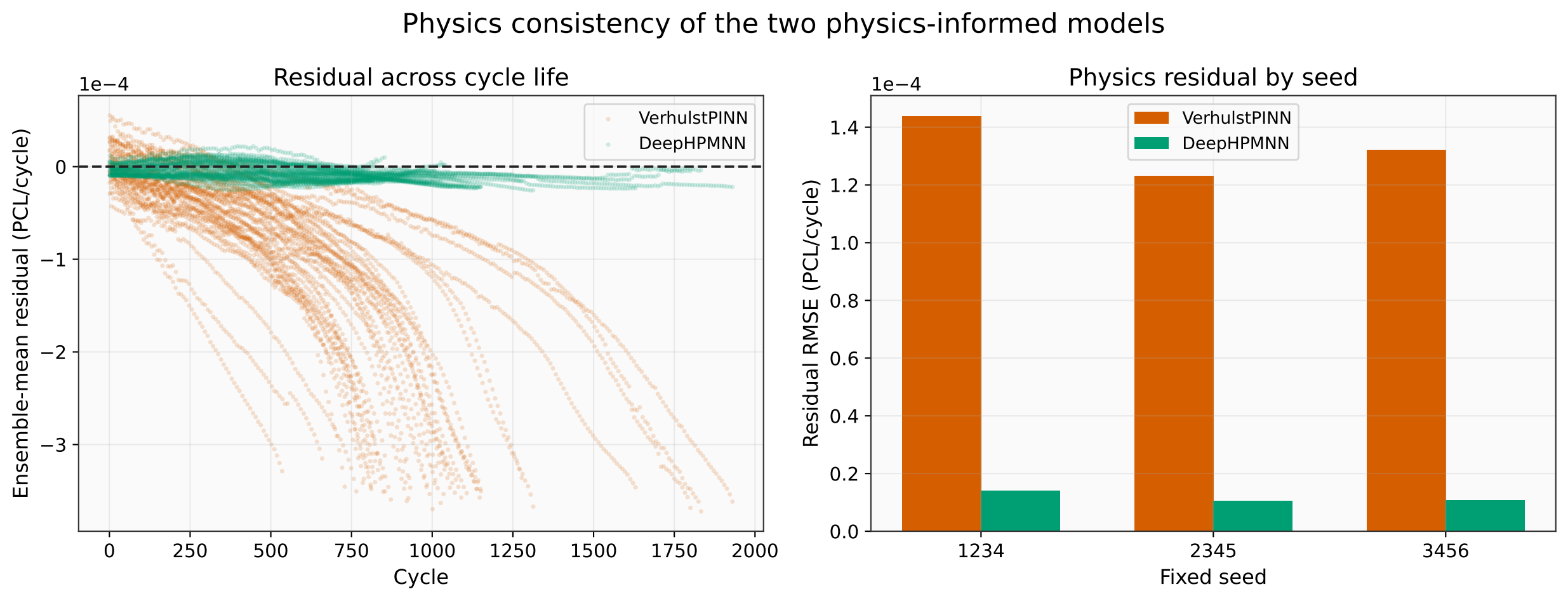

Physics consistency of the two physics-informed models
Residual across cycle life
1e−4
VerhulstPINN
DeepHPMNN
Ensemble-mean residual (PCL/cycle)
0
−1
−2
−3
0
250
500
750
1000
1250
1500
1750
2000
Cycle
Physics residual by seed
1e−4
VerhulstPINN
DeepHPMNN
Residual RMSE (PCL/cycle)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1234
2345
3456
Fixed seed